Kalainesan Natarajan

Computer Science Undergraduate | Backend Systems, LLM Agents, Python ML

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PROFESSIONAL OVERVIEW

Second-year B.Tech CSE student building backend systems, full-stack apps, LLM agents, and Python ML projects.

Skills: Python, JavaScript, TypeScript, SQL | React, Next.js | Node.js, Express.js, Socket.io | PostgreSQL, Supabase, MongoDB | LLM agents, scikit-learn, pandas | Git, Docker

Availability: Dubai, Dec 2026 (2-4 weeks), paid or unpaid. UAE resident. Remote open before, during, and after.

Offline Sync Conflict | Node.js, Express, PostgreSQL, Docker | github.com/kalainesan-n/offline-sync-conflict

- **Built a notes sync API that detects offline edit conflicts using base-version checks and field-level tracking, merging independent edits and keeping server values on overlaps.**
- **Used PostgreSQL transactions with SELECT FOR UPDATE and idempotency records for safe concurrent updates and retries. Added version restore, OpenAPI docs, Jest/Supertest tests, and Render deployment.**

CampusBorrow | Next.js, TypeScript, Supabase | campus-borrow-three.vercel.app | github.com/kalainesan-n/CampusBorrow

- **Built a peer-to-peer campus borrowing platform with auth, realtime updates, image uploads, and search. Enforced access control with PostgreSQL Row-Level Security.**
- **Prevented double claims with an atomic PostgreSQL RPC using row locking. Deployed on Vercel.**

BullWarden (TriAgent Advisory System) | OpenClaw, ArmorClaw (ArmorIQ), LLMs, Alpaca Paper API, React

- Built a three-agent paper-trading workflow: news sentiment analysis, a policy-based risk check, and execution of approved orders only (simulated trading, no real funds).
- Used ArmorClaw runtime policy enforcement to rate trades GREEN, YELLOW (size reduced), or RED (blocked), with spending limits and prompt-injection defences.
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Air Quality Forecasting | Python, pandas, scikit-learn
|https://github.com/kalainesan-n/AIML-Recruitment-2026-Kalainesan

- Built a one-hour-ahead CO forecasting pipeline on the UCI Air Quality dataset using lag, rolling-mean, and cyclic time features, shifted to prevent leakage.
- Trained Linear Regression with time-series cross-validation. It beat a persistence baseline on the holdout set (RMSE 0.694 vs 0.801, R^2 0.738 vs 0.651)

EDUCATION

B.Tech in Computer Science and Engineering, SRM Institute of Science and Technology, Kattankulathur, India | Expected 2029
GEMS Our Own High School, Al Warqa'a, Dubai | 2013-2023